6.2 Sequences of Transformations That Include Dilations

Lesson Introduction

Benchmarks	MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates. MA.912.GR.2.3 Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure.	Learning Targets	• I can describe a sequence of transformation that includes a dilation. • I can represent a sequence of transformations using coordinate notation with algebraic descriptors.

Benchmark Coherence	Building On	Working Toward
	MA.8.GR.2.2 Given a preimage and image generated by a single dilation, identify the scale factor that describes the relationship.	N/A

Essential Question	• How do you know if a sequence of transformations is commutative?
Lesson Narrative	In this lesson, students will be combining dilations with other transformations, such as translations and reflections. They will be working with a partner to write algebraic and written expressions for these combinations of transformations. Students will also investigate which transformations are commutative.

Component Summary				
	6.2.1 Warm-Up (~5 min.)	Determining an error with a dilation *(SE p. 291)*	6.2.3 Guided Practice (10 min.)	Writing an algebraic description for a sequence of transformations *(SE p. 294)*
	6.2.2 Guided Instruction (25-30 min.)	Describing a sequence of transformations *(SE p. 292)*	6.2.4 Your Turn! (5 min.)	Describing a sequence of transformations *(SE p. 294)*
	6.2.5 Wrap-Up (~5 min.)	Ticket Out the Door writing algebraic descriptions of transformations *(SE p. 295)*		

Resources	Glossary	Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> • Transformation • Dilation • Translation • Reflection • Preimage • Image	• (Optional) Graph Paper • (Optional) Cut Outs of Preimages and Images	• N/A

<table>
<tr><td rowspan="4">Teacher Tips</td><th>Common Misconceptions</th><th>Things to Consider</th></tr>
<tr><td>Students may not realize that some sequences of transformations are communitive and some are not.
Students may struggle getting the correct order of transformations to generate the same image from a preimage.</td><td>Consider spending a few moments reviewing the algebraic descriptions of the transformations learned in Unit 4 by encouraging students to locate those descriptions in their notes.
If time is an issue, then consider assigning 6.2.4 Your Turn! for homework.</td></tr>
<tr><th>Literacy and Language Routines</th><th>Mathematical Thinking and Reasoning (MTR) Standard</th></tr>
<tr><td>Think-Pair-Share
6.2.2 Guided Instruction is a natural place to implement this routine. The questions are designed to spark thinking for the individual learner first and then allow each student to make a conjecture about the answers to the questions. Then, students share with a partner to explain their reasoning, that can then be shared whole group.</td><td>MA.K12.MTR.5.1: Patterns and Structure
In 6.2.2 Guided Instruction, students discover additional methods for determining the scale factor of a dilation. Students use structure to understand and connect mathematical concepts and to focus on relevant details within a problem. They also create plans and procedures to logically order events, steps, or ideas to solve problems. This allows them to be able to decompose a complex problem into manageable parts.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">To provide a kinesthetic and visual element to the lesson, provide students with cut outs of the preimages and images in each of the components. Students can use the images to demonstrate the final position when following a sequence of transformations and when changing the order of the sequence to determine whether the transformations are commutative.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
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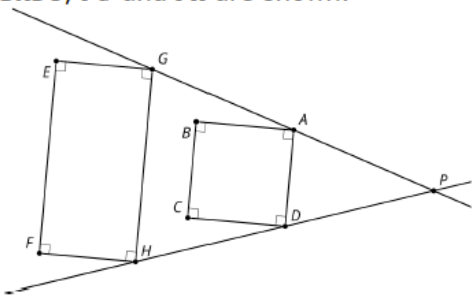
6.2.1 Warm-Up: Dilation Miscalculation

Component Overview: *Students determine if an image demonstrates the characteristics of a dilation and provide a justification of their reasoning.*

6.2.1 Warm-Up: Dilation Miscalculation

1. Cassandra tried to determine if the diagram shown represents a dilation.
   $EGHF$, $BADC$, $\overleftrightarrow{PG}$ and $\overleftrightarrow{PH}$ are shown.



Cassandra listed four characteristics of dilations:

Characteristic	Yes/No
Corresponding angle measures are equal	Yes
Corresponding sides are parallel	Yes
Corresponding vertices are collinear with the point of dilation	Yes
Orientation of the shapes are the same	Yes

a. Of the characteristics Cassandra listed, which are not true for $EGHF$ and $BADC$?
Corresponding vertices are collinear with the point of dilation.

b. Explain why the diagram shows that the dilation does not have these characteristics. If necessary, use the diagram to draw.
Not all corresponding vertices are collinear with the point of dilation. Points A and G are collinear with point P, as are points D and H. Points E and B are not collinear with point P and points F and C are not collinear with point P.

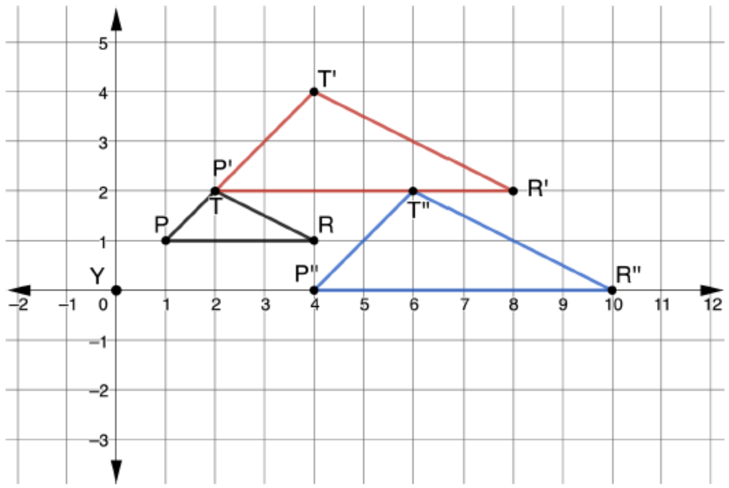
To add a level of complexity, have students make edits to the figures to demonstrate a true dilation.

6.2.2 Guided Instruction: Sequence of Transformations

***Component Overview**: Students will describe a sequence of transformations and then determine if the transformations are commutative.*

6.2.2 Guided Instruction: Sequence of Transformations

1. A sequence of transformations of ΔPTR is shown on the coordinate grid. ΔPTR is dilated then translated, where $\Delta P'T'R'$ is the dilation and $\Delta P''T''R''$ is the translation of $\Delta P'T'R'$.



ΔPTR	
P	$(1,1)$
T	$(2,2)$
R	$(4,1)$

$\Delta P'T'R'$	
P'	$(2,2)$
T'	$(4,4)$
R'	$(8,2)$

$\Delta P''T''R''$	
P''	$(4,0)$
T''	$(6,2)$
R''	$(10,0)$

a. Complete the description.

ΔPTR was dilated using a scale factor of _2__ centered at the origin. The resulting polygon, $\Delta P'T'R'$, was translated __*right 2, down 2*_ to create $\Delta P''T''R''$.

Think-Pair-Share
Consider employing the instructional routine Think-Pair-Share to allow students time to first think through each question, discuss their reasonings with their partner, and then bring their ideas to the whole group.

How can you determine if the scale factor will represent a ratio greater than one or less than one?

6.2.2 Guided Instruction: Sequence of Transformations (continued)

b. Complete the table.

Transformation	Algebraic Description
Dilation	$(x, y) \to (2x, 2y)$
Translation	$(x, y) \to (x + 2, y - 2)$

c. Determine the coordinates of each triangle if ΔPTR is first translated then dilated.

Pre-image

Triangle PTR	
P	$(1, 1)$
T	$(2, 2)$
R	$(4, 1)$

Translation

Triangle $P'T'R'$	
P'	$(3, -1)$
T'	$(4, 0)$
R'	$(6, -1)$

Dilation

Triangle $P''T''R''$	
P''	$(6, -2)$
T''	$(8, 0)$
R''	$(12, -2)$

Some sequences of transformations are commutative, which means the transformations can be applied in any order.

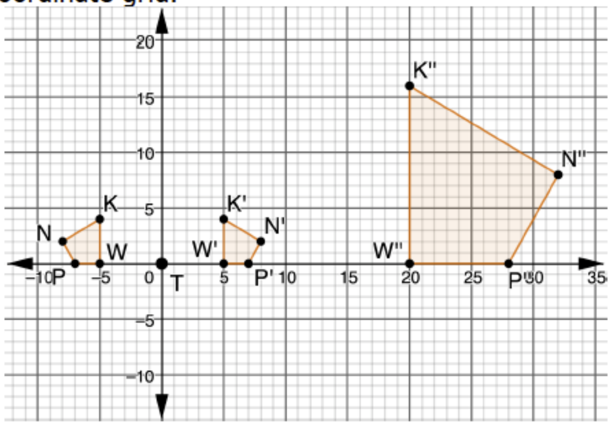
d. Use the tables to determine if the sequence of transformations applied to ΔPTR is commutative. Work with your partner to write an explanation.
The transformations are not commutative. When applying the sequence of transformations in a different order, the vertices for the two final images are not the same, so the images will not coincide.

Notice and Wonder
After completing question 1c, consider utilizing a Notice and Wonder routine to offer students the opportunity to make observations on when the transformations are applied in different orders.

Have students graph the points in the table of values for question 1c to have a visual representation of the resulting image when the transformations changed order.

6.2.2 Guided Instruction: Sequence of Transformations (continued)

2. $KWPN$, $K'W'P'N'$, and $K''W''P''N''$ are shown on the coordinate grid.



$KWPN$	
K	$(-5, 4)$
W	$(-5, 0)$
P	$(-7, 0)$
N	$(-8, 2)$

$K'W'P'N'$	
K'	$(5, 4)$
W'	$(5, 0)$
P'	$(7, 0)$
N'	$(8, 2)$

$K''W''P''N''$	
K	$(20, 16)$
W	$(20, 0)$
P	$(28, 0)$
N	$(32, 8)$

a. Discuss with your partner the first transformation that occurred. Describe how you determined the first transformation.
Answers will vary. The first transformation that occurred was a reflection. I know this because I see that the pre-image and image are reflected images of one another. I also know the algebraic description of a reflection over the y-axis is $(x, y) \rightarrow (-x, y)$.

b. Determine the scale factor in the second transformation.
The scale factor of the second transformation was 4.

Poll the Class
Consider using the instructional routine Poll the Class to begin a discussion on the transformation that occurred first. After the partner discussion, poll the class to determine what transformation occurred first. Call on different students to share their justifications for the transformation they chose.

c. Complete the description of the transformations.

$KWPN$ was _*reflected across the y-axis*_____ to create $K'W'P'N'$. $K'W'P'N'$ was dilated using a scale factor of _4_ centered at __Point T___ to create $K''W''P''N''$.

d. Complete the table.

Transformation	Algebraic Description
Reflection across y-axis	$(x, y) \rightarrow (-x, y)$
Dilation	$(x, y) \rightarrow (4x, 4y)$

e. Work with your partner to determine if the sequence of transformations applied to $KWPN$ is commutative. Complete the table by first applying the dilation, and then the other transformation. Then, write a conclusion based on the tables.

$KWPN$		$K'W'P'N'$		$K''W''P''N''$	
K	$(-5, 4)$	K'	$(-20, 16)$	K	$(20, 16)$
W	$(-5, 0)$	W'	$(-20, 0)$	W	$(20, 0)$
P	$(-7, 0)$	P'	$(-28, 0)$	P	$(28, 0)$
N	$(-8, 2)$	N'	$(-32, 8)$	N	$(32, 8)$

The sequence of transformations is commutative.

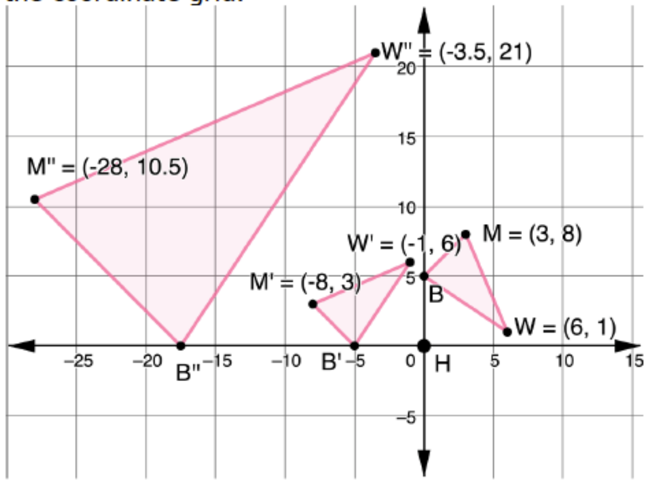
Provide opportunity for meaningful discussion as to if this sequence of transformations is commutative or not. When selecting a particular student to present their work, choose students who took different approaches to the problem. In fact, it may be advantageous to choose incorrect responses to highlight how and why they are incorrect. You can highlight a variety of responses or strategies or show a progression from simple to complex representation. Make sure that over time all students feel they are authors of the mathematical ideas presented.

6.2.3 Guided Practice: Mapping a Dilation

Component Overview: *Students will apply their knowledge of sequences of transformations by interpreting the graph, starting with the preimage, following through to the end of the mapping process, and writing the algebraic description.*

6.2.3 Guided Practice: Mapping a Dilation

1. A sequence of transformations of ΔMWB is shown on the coordinate grid.



a. Complete the table of values for each triangle.

Preimage **Dilation**

Triangle MWB	
M	$(3, 8)$
W	$(6, 1)$
B	$(0, 5)$

Triangle $M'W'B'$	
M'	$(-8, 3)$
W'	$(-1, 6)$
B'	$(-5, 0)$

Triangle $M''W''B''$	
M''	$(-28, 10.5)$
W''	$(-3.5, 21)$
B''	$(-17.5, 0)$

b. Complete the table.

Transformation	Algebraic Description
Rotation 90° counterclockwise	$(x, y) \rightarrow (-y, x)$
Dilation	$(x, y) \rightarrow (3.5x, 3.5y)$

Allow informal observation throughout the first two components to dictate how "guided" this Guided Practice should be for you and your classroom. Consider doing small-group instruction for some while others attempt independently.

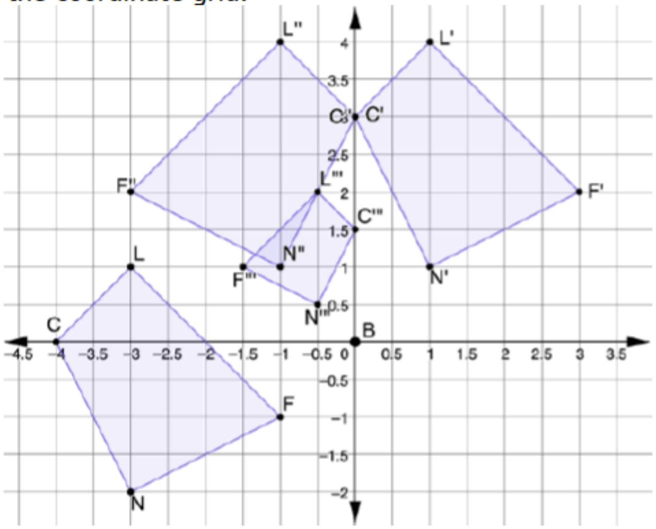
For students who finish early, challenge them to determine if the transformations in question 1 are commutative.

6.2.4 Your Turn!

***Component Overview**: Students will demonstrate their understanding of sequences of transformations by describing the transformations and then writing the algebraic descriptions.*

6.2.4 Your Turn!

1. $LFNC$, $L'F'N'C'$, $L''F''N''C''$, and $L'''F'''N'''C'''$ are shown on the coordinate grid.



   a. Describe the sequence of transformations of $LFNC$ that resulted in $L'''F'''N'''C'''$.
   $LFNC$ was shifted right 4 units and up 3 units to become quadrilateral $L'F'N'C'$. Then $L'F'N'C'$ was reflected across the y-axis to become $L''F''N''C''$. Finally, $L''F''N''C''$ was dilated from point B by a scale factor of $\frac{1}{2}$ to become $L'''F'''N'''C'''$.

   b. Write the algebraic descriptions of the sequence of transformations.

Transformation	Algebraic Description
Translation	$(x, y) \rightarrow (x + 4, y + 3)$
Reflection	$(x, y) \rightarrow (-x, y)$
Dilation	$(x, y) \rightarrow \left(\frac{1}{2}x, \frac{1}{2}y\right)$

Students may be more challenged now that they must apply and describe three transformations. Students might get stuck in getting the correct order of transformation to generate the same image from a preimage. The teacher can help the students by asking them to sketch their descriptions step-by-step using the algebraic description and see if they match the given image.

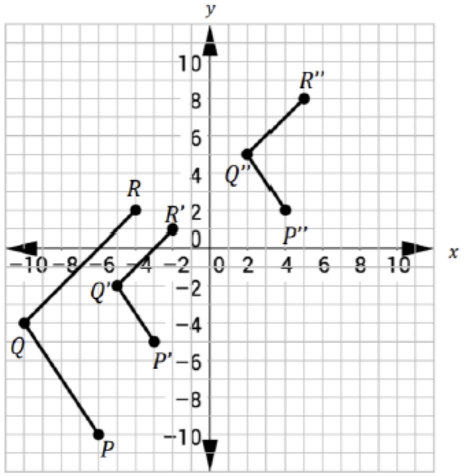
6.2.5 Wrap-Up: Ticket Out the Door

Component Overview: *6.2.5 Wrap-Up assesses students' abilities to describe the transformations and write an algebraic description.*

6.2.5 Wrap-Up: Ticket Out the Door

Consider using this exercise as a formative assessment to gather data on student progress for this lesson.

1. Points $P(-6,-10), Q(-10,-4),$ and $R(-4,2)$ are plotted on the coordinate grid to form figure PQR. Two transformations of figure PQR are shown.



Gladys first transformed figure PQR to $P'Q'R'$ using a dilation centered at the origin. She then transformed figure $P'Q'R'$ to $P''Q''R''$.
Complete the table.

Transformation	Written Description	Algebraic Description
PQR to $P'Q'R'$	*Dilation centered at the origin with a scale factor of $\frac{1}{2}$*	$(x,y) \rightarrow (\frac{1}{2}x, \frac{1}{2}y)$
$P'Q'R'$ to $P''Q''R''$	*Translation of 7 units up, 7 units to the right*	$(x,y) \rightarrow (x+7, y+7)$